Worksheet 1: Lipid Digestion — Concepts and Applications

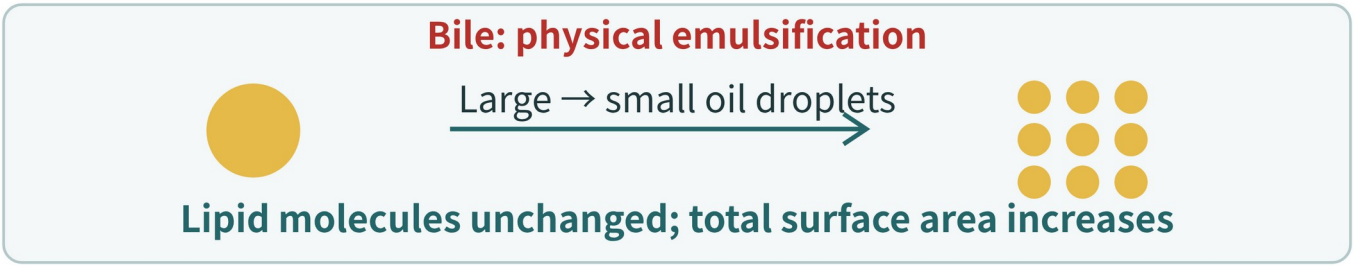
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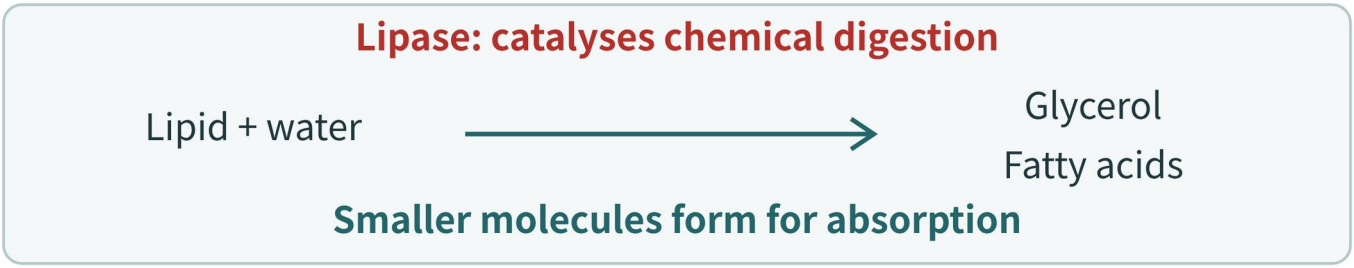
Read the topic notes first, then apply the concepts to analyse the results.

Topic notes

X is **bile**, made in the liver and stored in the gall bladder, containing no digestive enzymes. It **physically emulsifies** large oil droplets into smaller droplets without changing lipid molecules. The greater **total surface area** helps **lipase** contact lipids.

Y is **pancreatic juice** containing **lipase**. **Lipase** catalyses **chemical digestion**, forming **glycerol** and **fatty acids** in this course's simplified complete-hydrolysis model. **Lipase** is a **protein**: boiling **denatures** it and stops its action, which cooling does not restore. **Bile** retains emulsifying action after boiling in this model.





Context: lipids need digestion before absorption. What happens after adding digestive juice X or Y?



Part 1 | Context and fair comparisons

Researchers added 2 mL X or 2 mL Y to identical oil–water mixtures, shook them identically at 37°C, and compared appearances after one hour.

1. Identify the variables: choose all appropriate factors in each row.

IV(the factor changed on purpose)；DV(the result you measure or observe)；CV(the factors kept the same)

Independent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ **Combination of digestive juices**
Dependent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ **Appearance of the mixture** ／ Combination of digestive juices
Controlled variables	**Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking** ／ Appearance of the mixture ／ Combination of digestive juices

2. Select the assumptions for a fair comparison (choose all that apply)

(An assumption is a condition we have not proved but take as true, to help make the experiment results valid.)

☑ **Observe all setups after the same reaction time.**

☑ **Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.**

☑ **Use the same method of shaking for every setup.**

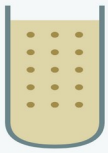
□ The X setup may receive more total liquid than the Y setup.

Part 1 | Apply concepts to the individual actions of X and Y

Both model results show one hour after adding the juice and shaking.

Set X

Add 2 mL X



Cloudy emulsion

Set Y

Add 2 mL Y



Thinner oil layer; two layers remain

3. Appearance of X after one hour

☑ **a cloudy emulsion**

☐ The oil layer disappears

4. Appearance of Y after one hour

☑ **Thinner oil layer, with two layers remaining**

☐ No obvious change in the oil layer

Apply emulsification and surface-area concepts to analyse how X helps Y.

Researchers used two fresh tubes, added X or equal water and shook them, then added the same amount of Y. Pretreatment times were equal.

Part 2 | Extension 1: how does X help Y?

The following model results show one hour after adding Y.

X first, then Y

1 mL X → 1 mL Y



Less cloudy
No obvious oil layer

Water first, then Y

1 mL water → 1 mL Y



Thinner oil layer
Two layers remain

5. How does X help Y act?

X disperses large oil droplets into smaller droplets. The total surface area of the same amount of lipid:

☑ **increase**

☐ decrease

Part 2 | Extension 1: further analysis

6. What does appearance tell us?

☑ **Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol and fatty acids have formed.**

□ Disappearance of the oil layer completely proves that lipids have broken down into glycerol and fatty acids.

Part 3 | Extension 2: effects of boiling

Researchers used unboiled X followed by Y as the baseline. They boiled one juice, cooled it to 37°C, then added 1 mL each of X and Y in the same order; results show one hour later.

Boiled X + unboiled Y

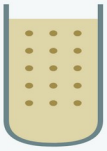
Use after cooling



Less cloudy; no oil layer

Unboiled X + boiled Y

Use after cooling



Cloudy emulsion

7. Which juice loses its action after boiling?

□ Digestive juice X

☑ **Digestive juice Y**

Integrated explanation

8. Using the experimental results and topic notes, explain how bile and lipase in pancreatic juice work together to help digest lipids, and how boiling affects their actions.

Bile physically emulsifies large droplets into small droplets without changing lipid molecules. The greater total surface area helps lipase act. Lipase in pancreatic juice catalyses chemical digestion into glycerol and fatty acids in this model. Lipase is a protein; boiling denatures it and stops its catalytic action even after cooling. Bile retains its emulsifying action after boiling.